Dhruv Dilipkumar Prajapati

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Mechanical Engineering Junior student with expertise in design, CAD modeling, and finite element analysis (FEA), passionate about semiconductor, automotive and aerospace industries. Skilled in data analysis, test & validation, Modelling and optimisation, Six Sigma approaches and Lean principles, seeking a summer and/or fall 2026 co-op/internship blocks.

EDUCATION

Bachelor of Science in Mechanical Engineering, Rochester Institute of Technology, NY – 14623, USA, GPA: 3.30 **Aug 2027**

Coursework: Systems Dynamics & Modeling, Circuits, Heat Transfer, Intro to Acoustics, Dynamics, Material Science, Semiconductor Innovation & Manufacturing, EUV Photolithography Technology, Fluid Mechanics, Thermodynamics, Statics, Strength of Materials, Engineering Measurements lab, Computer-Aided Design/Manufacturing, Multivariable Calculus, Ordinary Differential Equations, Boundary-value Problems.

CORE COMPETENCES

- ***CAD Design & Analysis***: SolidWorks, Flow Simulation CFD, ACAD 2D/3D, Ansys Mechanical FEA, CATIA v5, Prusa i3, GD&T
- ***Programming***: Python, MATLAB, Simulink, CFTool, MySQL (intermediate), LabVIEW
- ***Tools & Fabrication***: Lathe, CNC, Surface Finish, Shop Machine tools, Prusa i3 machine, Instron machine, Quasar 10 Tensile Tester, Rockwell Tester
- ***Office***: Microsoft 365, Excel, PowerPoint, Outlook, Power BI (basic), SAP

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant **February 2025 – Ongoing**
Fluids Acoustics STructures Laboratory (FAST Lab) **Rochester, NY**

- Topic: Vortex ring impingement on concave cavities - applies to speech production, enhanced heat transfer in slot fin cooling & variety of other flow scenarios.

Mechanical Engineering Intern **Aug 2025 – Dec 2025**
MKS Instruments **Rochester, NY**

- Performed SolidWorks Flow Simulation on a pulsed DC heat sink design for semiconductor fabrication and validated results through lab testing (heat sink, chiller, flow meter, pressure gauges), compared experimental and simulated pressure drop data to support new product development at MKS.
- Owned Volumetric flow rate & fluid modelling analysis project for the R&D dept and increased the efficiency of tests by almost 30%, and reduced the cost by $6000.
- Supported requalification of technical instruments & work stations using SolidWorks CAD design, and updated important information on 2D drawings using ACAD and validated Bill of Materials (BOM) with modified designs.
- Worked with the Reliability group at MKS on a cross-functional project and helped in LabVIEW set-up for the newly installed ultrasonic clamp-on flowmeters for burn-in hardware testing, directly supporting DFx principles.
- Participated in the dept. projects involving HVAC controls, boilers and chillers, compressors, heat pumps, computer networks, bulk liquid nitrogen systems, and environmental ALT HASS/ESS chambers for reliability testing
- Regularly communicated and worked with external vendors on parts requirements, customer specifications, & design properties to manufacture small mounts and fixtures for the TechOps group for circuit testing.

Undergraduate Teaching Assistant **Aug 2024 – July 2025**
Rochester Institute of Technology **Rochester, NY**

ACADEMIC PROJECTS

Quarter Car Suspension Model – Simulink Project **Jan 2026 – Feb 2026**

- Created a Simulink model of the quarter-car suspension behaviour using defined parameters & simulated the response of the suspension going up onto a curb with a spring-mass-damping system.

Pulsed DC – Heat Sink Differential Pressure Project **Aug 2025 –Dec 2025**

- Performed SolidWorks Flow Simulation on Pulsed DC heat sink design for semiconductor fabrication and validated results through lab testing (heat sink, chiller, flow meter, pressure gauge), comparing. Experimental and simulated pressure drop data to support new product development at MKS.

MATLAB Lunar Lander Project **Oct 2024 – Dec 2024**

- Developed MATLAB simulation of lunar lander descent, modelling gravity, thrust, and fuel consumption and optimised code to achieve an accurate landing trajectory within a 5% error margin.

Reverse Engineering Project **April 2024 – May 2024**

- Directed team in disassembling and redesigning silver-armoured connectors using SolidWorks and fabricated a prototype via 3D printing, enhancing original design efficiency by 10%.